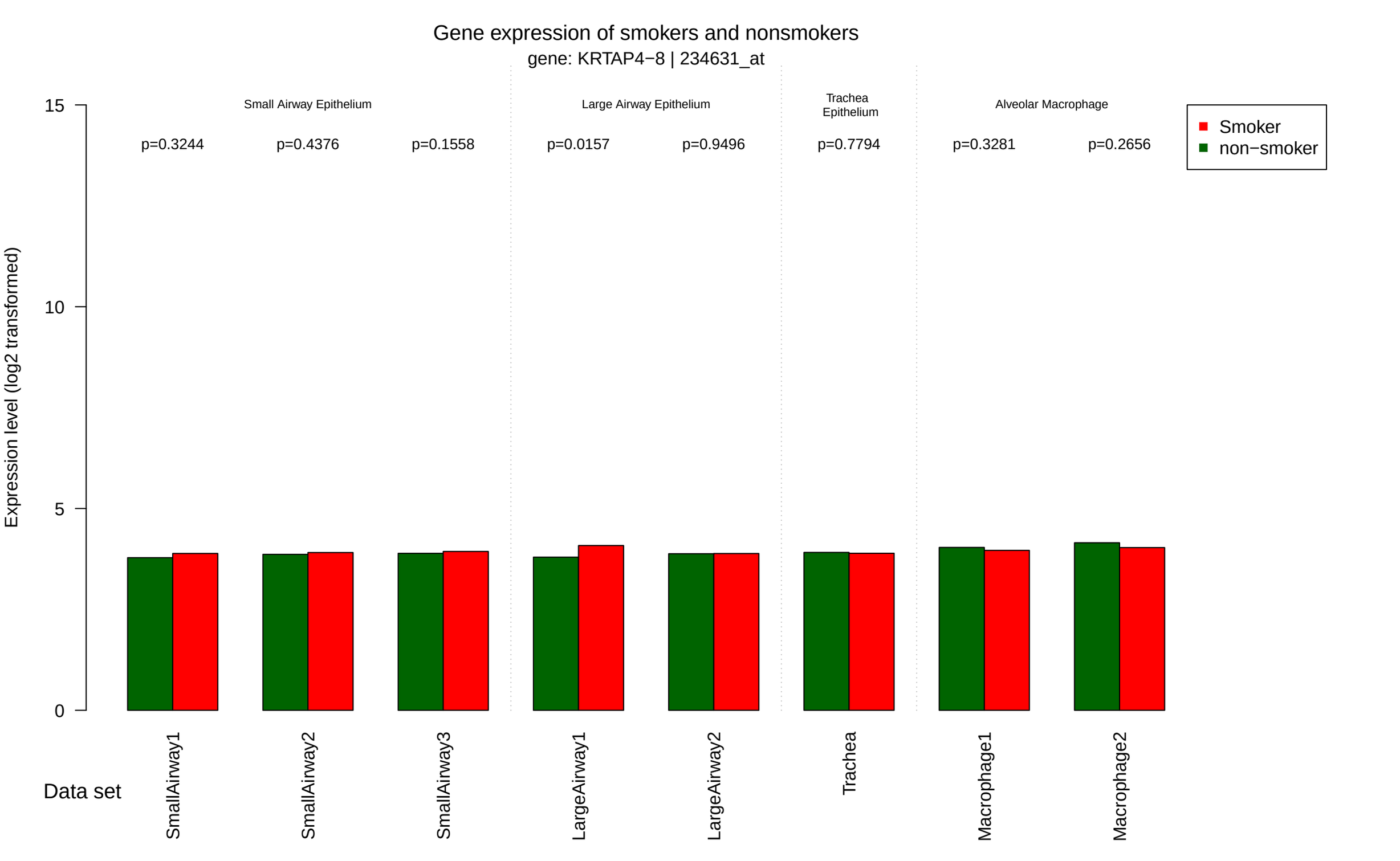
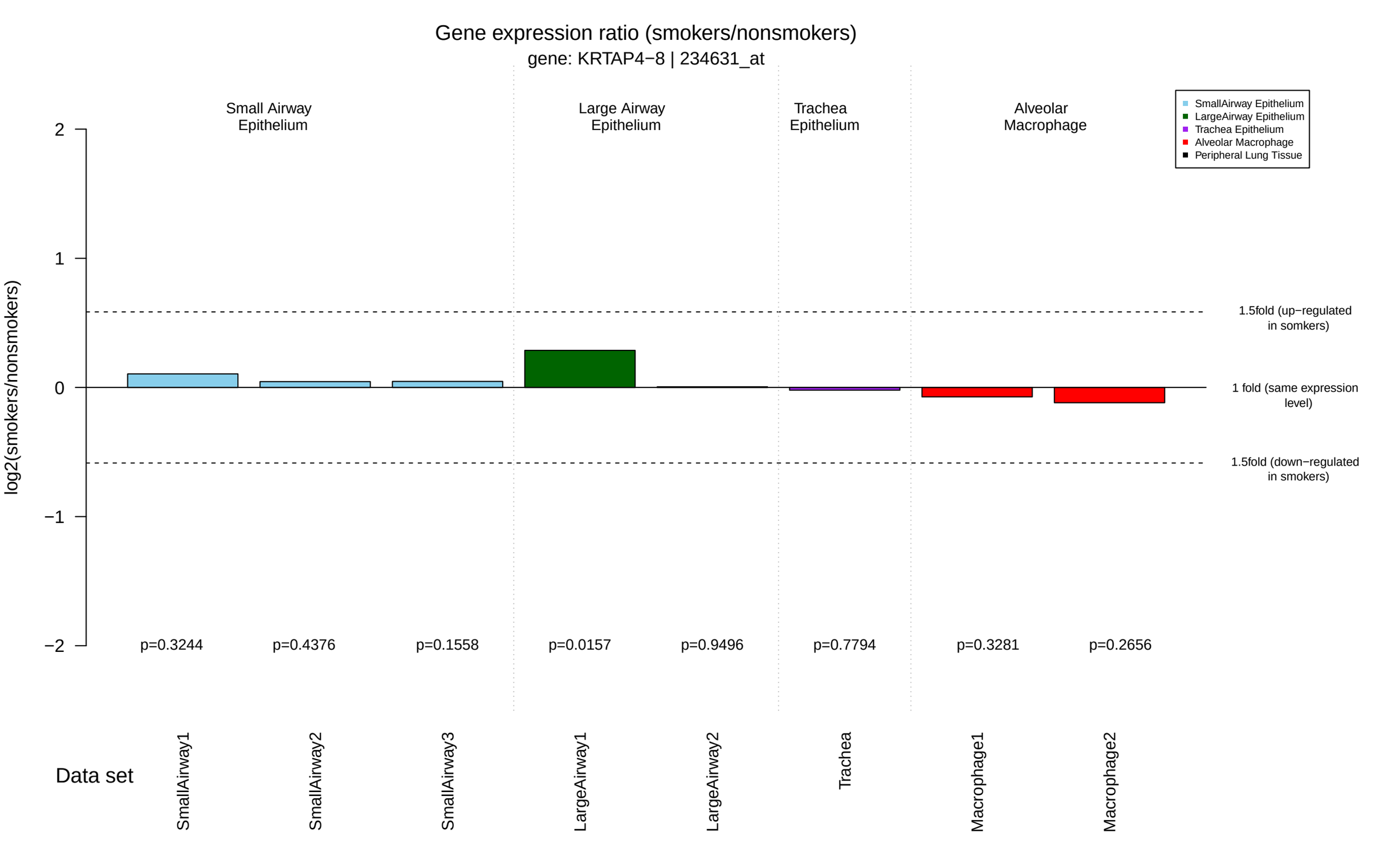
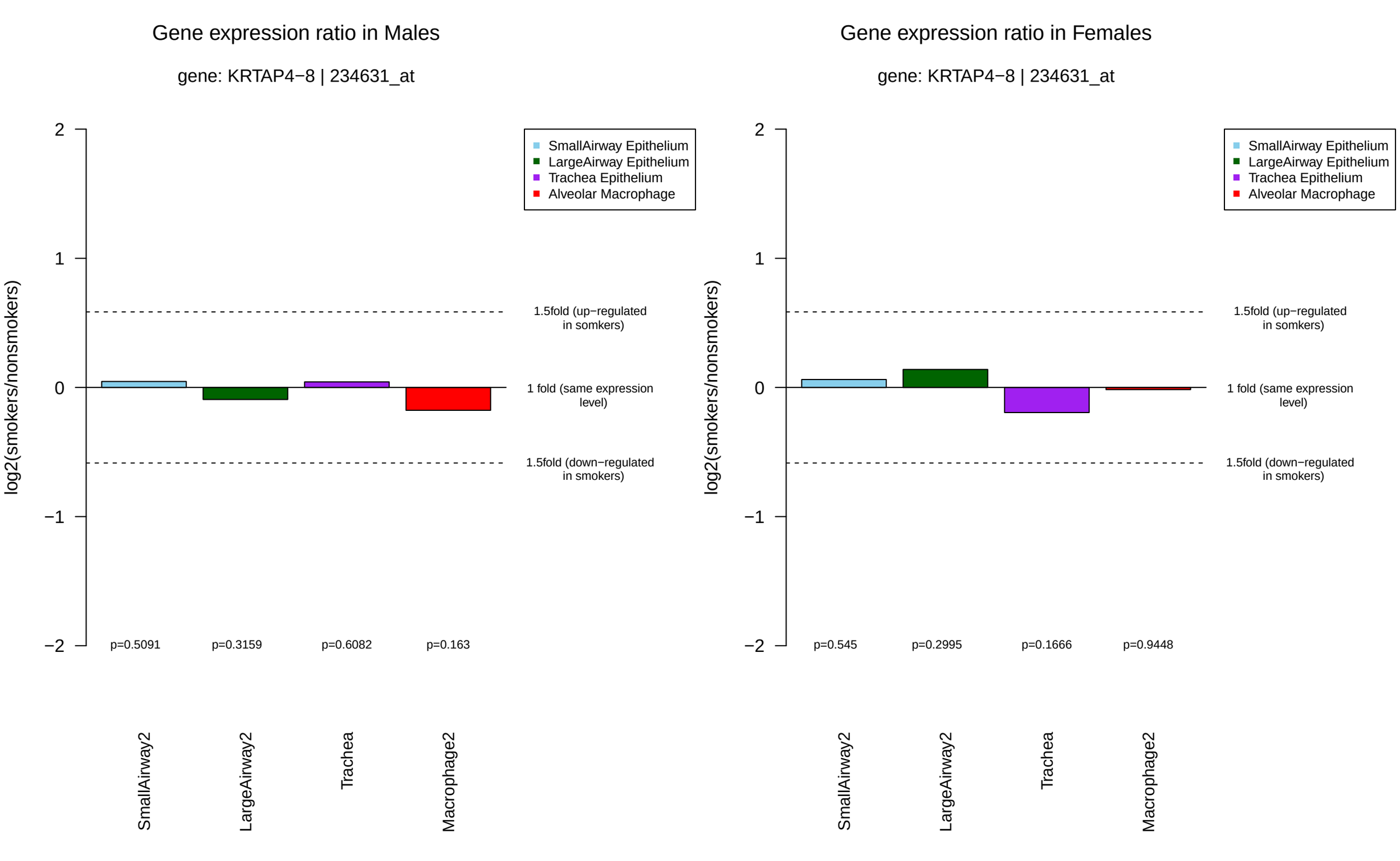
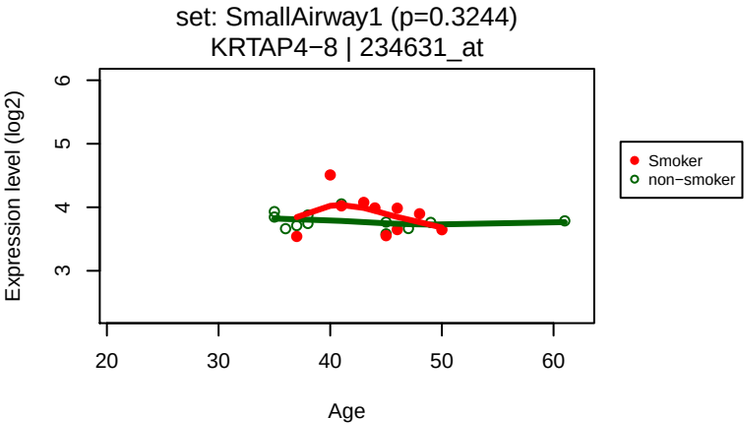
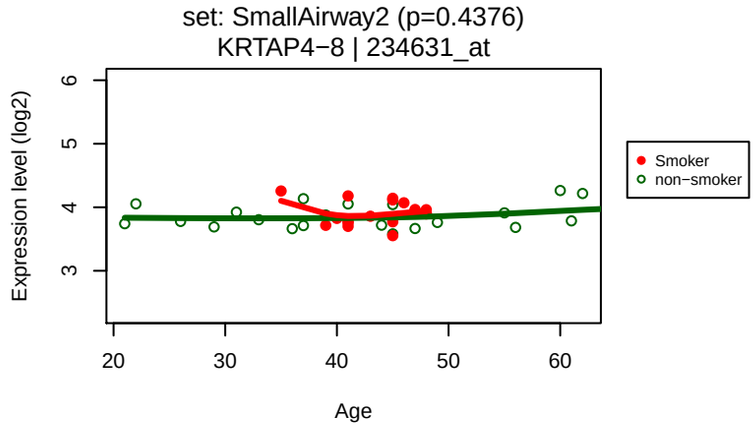
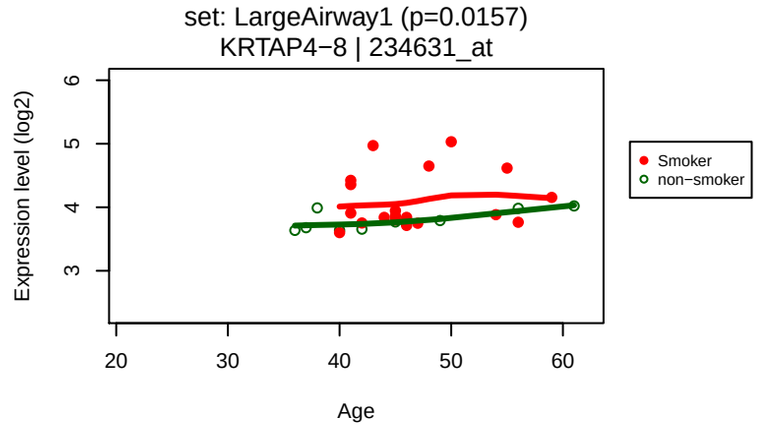

Gene expression of smokers and nonsmokers
gene: KRTAP4−8 | 234631_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3244
p=0.4376
p=0.1558
p=0.0157
p=0.9496
p=0.7794
p=0.3281
p=0.2656
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

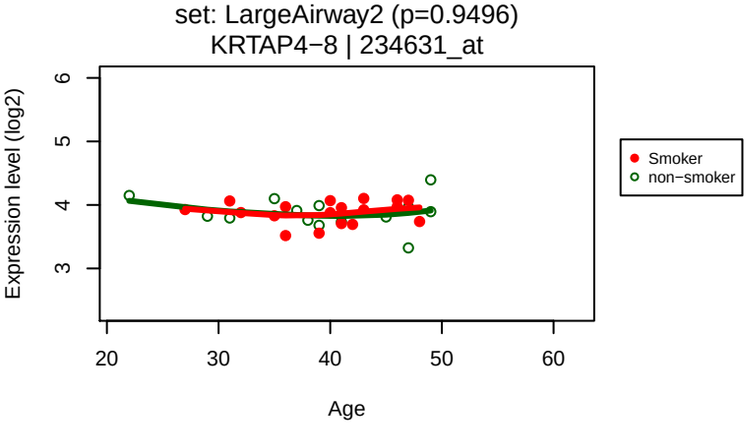

Gene expression ratio (smokers/nonsmokers)
gene: KRTAP4−8 | 234631_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3244
p=0.4376
p=0.1558
p=0.0157
p=0.9496
p=0.7794
p=0.3281
p=0.2656
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

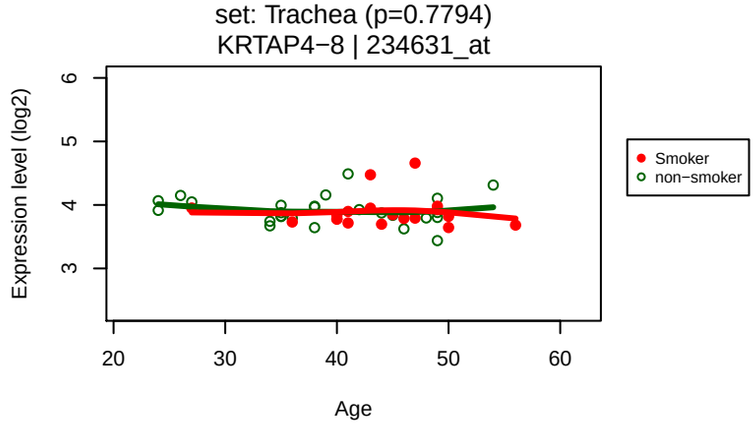

Gene expression ratio in Males
gene: KRTAP4−8 | 234631_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5091
p=0.3159
p=0.6082
p=0.163
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KRTAP4−8 | 234631_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.545
p=0.2995
p=0.1666
p=0.9448
SmallAirway2
LargeAirway2
Trachea
Macrophage2

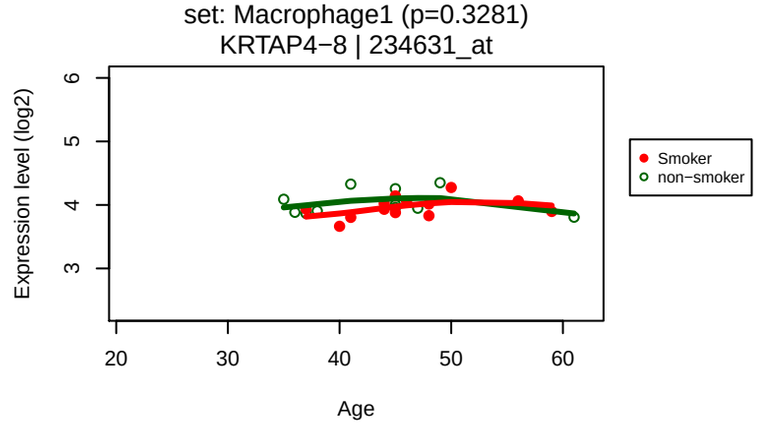

set: SmallAirway1 (p=0.3244)
KRTAP4−8 | 234631_at
Expression level (log2)
Age
Smoker
non−smoker

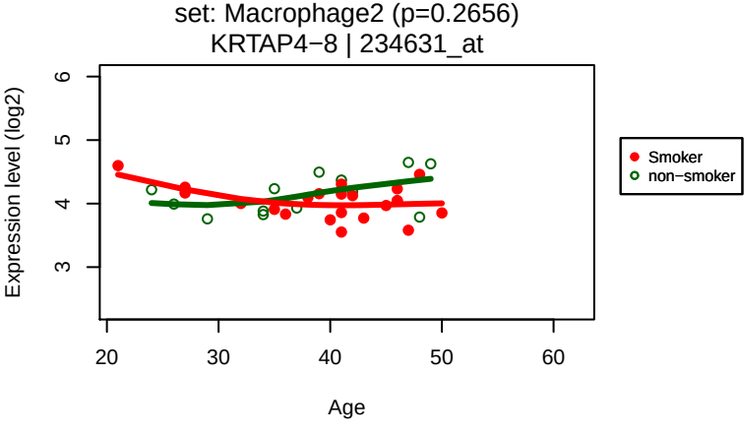

set: SmallAirway2 (p=0.4376)
KRTAP4−8 | 234631_at
Expression level (log2)
Age
Smoker
non−smoker

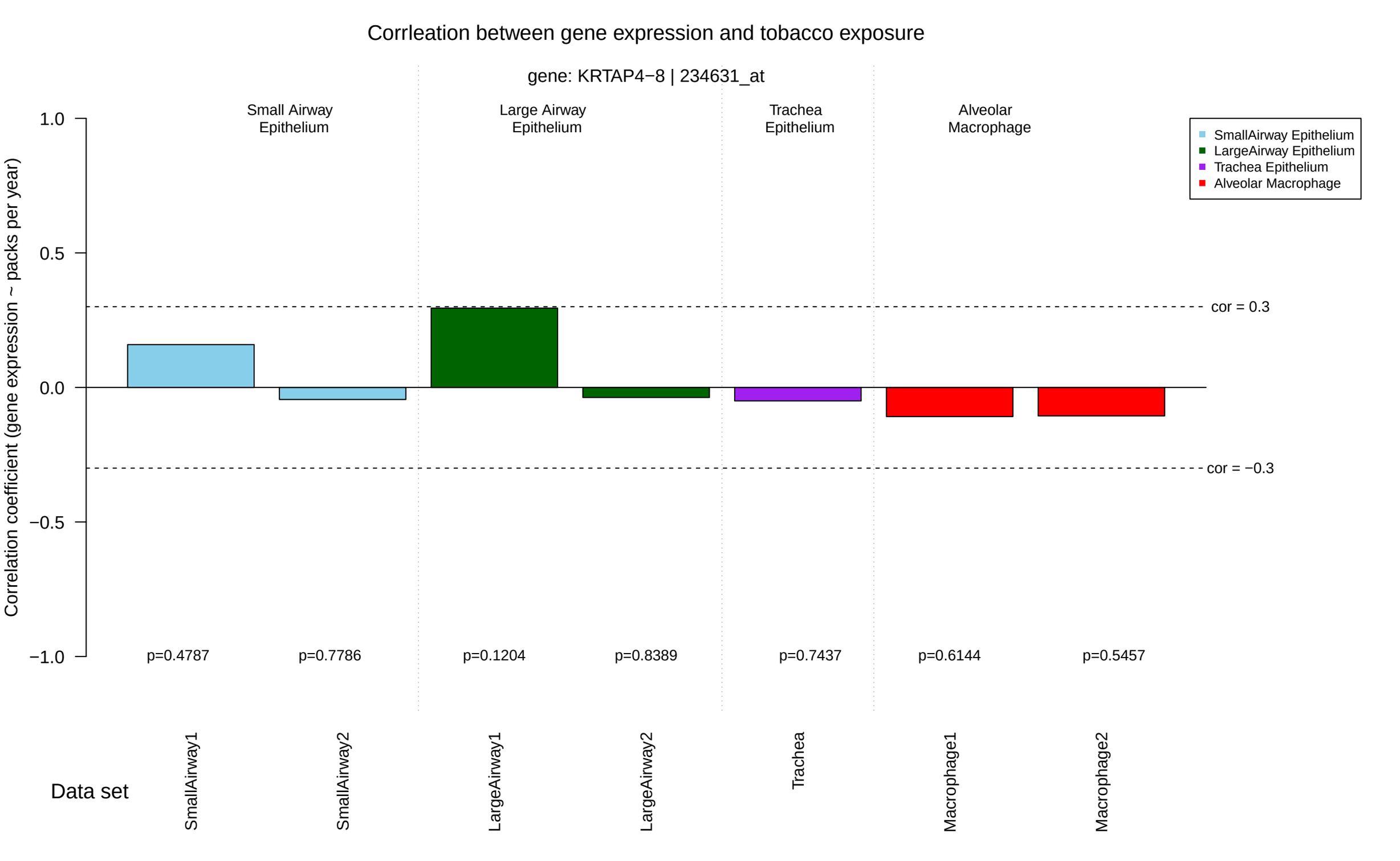

set: LargeAirway1 (p=0.0157)
KRTAP4−8 | 234631_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9496)
KRTAP4−8 | 234631_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7794)
KRTAP4−8 | 234631_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3281)
KRTAP4−8 | 234631_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2656)
KRTAP4−8 | 234631_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KRTAP4−8 | 234631_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4787
p=0.7786
p=0.1204
p=0.8389
p=0.7437
p=0.6144
p=0.5457
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2